MILITARY SPECIFICATION

LINER MATERIAL, POLYBUTADIENE BASE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of polybutadiene base liner material.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-509	Plastic Sheet, Rod and Tube, Laminated, Thermosetting
TT-D-301	Dibutyl Phthalate: Plasticizer (For Use in Organic Coatings)

MILITARY

MIL-D-6998	Dichloromethane, Technical
MIL-P-23942	Polybutadiene, Linear, Carboxyl Terminated
MIL-A-81236	Adhesive: Epoxy Resin with Polyamide Curing Agent
MIL-R-82657	Resin, Epoxy, Trifunctional

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1377

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DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CODE IDENT 30003)

673AS1128	Modifier, Cure
673AS1130	Curative
673AS1131	Filler, Liner
736AS1115	Tri-Phosphite

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the Government procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 412-75	Tension Testing of Rubber
ASTM D 792-66	Specific Gravity and Density of Plastics by Displacement

(Applications for copies should be addressed to the American Society for testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

CHEMICAL PROPULSION INFORMATION AGENCY (CPIA)

CPIA Publication No. 21	ICRPG Solid Propellant Mechanical Behavior Manual

(Application for copies should be addressed to the Chemical Propulsion Information Agency, Johns Hopkins University, Applied Physics Laboratory, 8621 Georgia Avenue, Silver Spring, MD 20910.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

REQUIREMENTS

3.1 Preproduction sample. When specified in the contract (see 6.2), the contractor shall produce a preproduction sample of the liner material for preproduction inspection and approval (see 4.4.1 and 6.2). The sample shall be manufactured using the methods and procedures proposed for

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production. The sample shall be tested as specified in section 4 for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Formulation. The formulation of the liner material shall be in accordance with TABLE I and the following:

TABLE I. Liner Material formulation.

	Ingredients	Specification or drawing	Parts by weight
Liner Material	Binder:		
	Polybutadiene	MIL-P-23942, type II	66.88
	Curative	673AS130	1.60 to 2.65
	Resin, liquid, epoxy	MIL-R-82657	1.15 to 1.90
	Dibutyl phthalate	TT-D-301	8.36
	Tri-phosphite	736AS115	1.67
	Dry:		
	Liner filler	673AS131	16.72
	Cure modifier	673AS128	2.68
	Diluent:		
	Dichloromethane	MIL-D-6998, grade B	Approx. 1.5/1.0 diluent to liner material

3.2.1 Curative and liquid epoxy resin. The amount of curative and liquid epoxy resin shall be adjusted within the range specified in TABLE I so as to provide the cured liner physical properties specified in TABLE II.

3.2.2 Diluent. The ratio of diluent to liner material shall be adjusted from batch to batch so as to provide the proper viscosity for application of the liner material and to result in the proper liner weight.

3.2.3 Weighing accuracy. Each liner ingredient shall be weighed to an accuracy of 0.7% of the values specified in TABLE I or to the nearest 0.1 gram (g), whichever is greater.

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3.3 Cured liner physical properties. The liner shall be cured in two steps. The precure shall be accomplished by exposure to 74 ± 5.5°C (165 ± 10°F) for 40 to 64 hours (h). The final cure of the liner shall be obtained during the cure of the applicable propellant. The physical properties of the final cured liner shall be in accordance with TABLE II.

TABLE II. Cured liner physical properties

Property	Test temperature		
Liner Bond to Propellant	77°F		165°F
(a) * 24 Hours - Maximum tensile strength, minimum (psi)	80		51
(b) ** 24 Hours - Type Failure	CP or CPI		
(a) * 4 Weeks - Maximum tensile strength, minimum (psi)	90		64
(b) ** 4 Weeks - Type failure	CP or CPI		
(a) * 8 Weeks - Maximum tensile strength, minimum (psi)	90		64
(b) ** 8 Weeks - Type failure	CP or CPI		
(a) * 13 Weeks - Maximum tensile strength, minimum (psi)	91		57
(b) ** 13 Weeks - Type failure	CP or CPI		
Liner	-65°F	77°F	165°F
(c) Maximum stress, minimum (psi)	300	100	75
(d) Elongation at maximum stress, minimum (%)	175	100	75
(e) Density (g/cm^3), 68 to 86°F	0.941 to 1.052		

* Storage conditions for all samples tested shall be 165°F at 22 percent relative humidity. One half of samples from each withdrawal shall be tested at 77°F and one half shall be tested at 165°F.

** Refer to CPIA Publication No. 21, Section 4.7.1 - 3.

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3.4 Workmanship. The liner material shall be uniform in consistency, free from contamination, foreign materials, or any other defect that would prevent its use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. Preproduction inspection (see 4.5).
2. Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

4.4 Sampling.

4.4.1 Preproduction sampling. When specified in the contract (see 6.2), a preproduction sample batch shall be prepared as specified in 3.1 and subjected to the preproduction inspection specified in 4.5.

4.4.2 Quality conformance inspection sampling. The quality conformance inspection consists of in-process monitoring of the weighing and blending of constituents (see 4.6). Sampling, therefore, shall not be required.

4.4.3 Batch. A batch shall consist of all liner material produced at one time in one mixer.

4.5 Preproduction inspection. The preproduction inspection shall consist of the tests of 4.5.3 through 4.5.5 conducted on specimens prepared in accordance with 4.5.1 and 4.5.2 from the sample liner material of 4.4.1.

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4.5.1 Liner bond specimen preparation. The liner bond specimens shall be prepared as follows:

a. Clean and line to the proper thickness one side of each of two 6 X 6 X 0.031 inch thick glass epoxy sheets, conforming to grade G10, type IV of L-P-509, concurrent with the cleaning and lining of the applicable inhibitor tubes. Precure at 74 ± 5.5°C (165 ± 10°F) for 40 to 64 h.

b. Remove sufficient mixed propellant from the propellant container after mixing and prior to casting the propellant grains to make propellant layer 0.5 inches thick for a. above. Place this uncured propellant between the two lined sheets.

c. Place the resultant sandwich under a minimum vacuum of 27 inches of mercury for 1 minute and press the sandwich with a force of 500 to 1,000 pounds for 15 to 30 seconds while under vacuum. The pressing shall be done in a close fitting square mold.

d. Cure the liner/propellant sandwich in its mold in the cure oven concurrent with the batch of propellant grains it represents.

e. Cut the 6 X 6 inch sandwich into 1 X 1 inch pieces. Sand the external flat surfaces of the glass epoxy sheet with course sandpaper (approximately 60 mesh), and bond on the mild steel tees with a 50/50 mix by weight of adhesive and curing agent in accordance with MIL-A-81236.

f. Repeat the procedure for an additional two glass epoxy sheets.

4.5.2 Cured liner material specimen preparation. Sample liner material shall be formed into sheets and precured in a platen press at 74 ± 5.5°C (165 ± 10°F) for 40 to 64 h. The samples shall then be cured in the cure oven concurrent with the batch of propellant grains they represent.

4.5.3 Liner bond test. The test specimens of 4.5.1 shall be tested in accordance with CPIA publication Number 21 to verify conformance with TABLE II (a) and (b). Five specimens shall be tested for each time-temperature combination. Failure of the average bond strength of the five specimens or of the type of failure of any specimen to meet the requirements of TABLE II (a) and (b), respectively, shall be cause for rejection of the preproduction sample.

4.5.4 Stress and elongation test. The maximum stress and the elongation at maximum stress of five specimens from the sample of 4.5.2 for each temperature specified in TABLE II (c) and (d) shall be determined in accordance with ASTM D 412-75, method A, die C. Failure of the average value of the five specimens to conform to the requirement of TABLE II (c) or (d) shall be cause for rejection of the preproduction sample.

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4.5.5 Density test. Three test specimens of 2 to 10g from the sample of 4.5.2 shall be subjected to a density test in accordance with ASTM D 792-66, method A-1 with the exceptions that isopropyl alchohol be used in place of water and that the test temperature be 20 to 30°C (68 to 86°F). Failure of the average value of the three specimens to conform to the requirement of TABLE II (e) shall be cause for rejection of the preproduction sample.

4.6 Quality conformance inspection. The quality conformance inspection shall consist of the inspection of weighing and blending procedures and techniques. The contractor shall maintain records consisting of mix sheets for each batch of liner material produced. The mix sheets shall be available for inspection by the Government upon request.

5. PACKAGING
This section is not applicable to this specification.

6. NOTES

6.1 Intended use. Liner material in accordance with this specification is intended to be used as a liner between the motor tube and the propellant of various cartridge and propellant activated devices; specifically, the M119 rocket motor assembly in accordance with MIL-R-50753.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Quantity required.
c. Applicable drawings for motor tube, liner material thickness and type of propellant.
d. When a preproduction sample is required (see 3.1, 6.3 and 6.4).
e. Inspection conditions when other than as specified (see 4.5).

6.2.2 Contract data requirements. Any data required for delivery in connection with this document shall be specified on a DD Form 1423 incorporated into the contract. Such data will be delivered as identified on completed (numbered) DIDs (Data Item Descriptions/DD 1664) which will be documented in the Applicable ADL (Authorized Data List).

6.3 Preproduction sample. When a preproduction sample is required (see 6.2d), it shall be tested and approved under the appropriate provisions of 7-104.55 of The Armed Services Procurement Regulation (ASPR). The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the preproduction sample.

6.4 Preproduction sample waiver. Preproduction samples submitted and approved on a recent contract (a contract remaining in production or a contract out of production for less than one year) may be accepted by the procuring activity in lieu of an additional preproduction inspection.

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When the production sample is waived (see 6.2.1 d.), the procurement document should contain a statement specifying that the standards of workmanship exhibited by the previously approved preproduction sample shall determine the minumum requirements of the current contract or order.

6.5 Suggested source of supply. A product that has met the requirements of this specification in past procurement actions is UPCO 44-055, Universal Propulsion Co., P. O. Box 3126, 117 Hayden Rd., Tempe, AZ 85281. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-L-82677(OS), "Liner Material, Polybutadiene Base"

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